A SYSTEMATIC APPROACH TO THE RIEMANN-ROCH THEOREM

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Abstract. This paper develops a systematic proof of the Riemann-Roch theorem for smooth proper curves. We begin with the classical language of divisors and spaces of rational functions, and then reinterpret divisors through line bundles and the canonical bundle of Kähler differentials. After introducing sheaf cohomology, we prove an Euler characteristic form of Riemann-Roch. We then describe first cohomology using principal parts and prove Serre duality through residues, trace maps, and reduction to vector bundles on the projective line. Combining these results yields the full Riemann-Roch theorem, from which we deduce the degree of the canonical divisor, the Riemann-Hurwitz formula, and the agreement of the algebraic and topological notions of genus.

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1. Introduction

The Riemann-Roch theorem has numerous formulations in many fields of mathematics. In a general sense, Riemann-Roch theorems relate some topological information of a space, like the genus, to algebraic information, such as the dimension of the vector space of functions on the space with some constraints on their zeros and poles. This type of idea turns out to be extremely important in many fields, including number theory, Riemannian geometry, and algebraic geometry. This paper is intended for a reader who has seen classical algebraic curves and is willing to accept some standard facts about schemes, sheaves, and cohomology. The goal is to explain how the modern proof of Riemann-Roch is organized. The reader will be introduced to tools such as line bundles and cohomology, allowing for a systematic and motivated proof of the algebraic Riemann-Roch theorem.

2. The Language of Divisors

In the following sections, I will prove one of the lower hanging results in the style of a Riemann-Roch theorem. This will illuminate the general flavour of these types of theorems. I will assume that the reader has an understanding of the language of classical algebraic geometry, point set topology, and commutative algebra.

For the remainder F will denote a nonsingular projective plane curve and k an algebraically closed field of characteristic 0.

Definition 2.1 (Divisors)**.** We denote $\operatorname{Div} F$ as the free abelian group generated by points of F. Therefore a *divisor* $D \in \operatorname{Div} F$ is given by a formal linear combination of finitely many points of F

$$D = a_1 P_1 + \cdots + a_n P_n \qquad P_i \in F \qquad a_i \in \mathbb{Z}$$

with addition of divisors given pointwise. We call the coefficient in front of some $P \in F$ the *multiplicity* of D at P. The degree of a divisor is given by the formula

$$\deg D = a_1 + \cdots + a_n$$

which counts how many points the divisor has with multiplicity. We can define a partial order on divisors by $D_1 \geq D_2$ if at every point the multiplicity of D_1 is greater than or equal to the multiplicity of D_2. We call a divisor *effective* if $D \geq 0$, i.e. it has nonnegative multiplicity at every point.

Construction 2.2 (Divisor of a Rational Function)**.** Consider a rational function φ in the function field $K(F)$ of F. In this context the rational functions are exactly those functions which can be written as $\varphi = f/g$ for $f, g \in S_d(F)$, where $S_d(F)$ denotes the *homogeneous polynomials* of degree d. In words, the rational functions are quotients of homogeneous polynomials of the same degree.

Since F is assumed to be nonsingular, the local ring $\mathcal{O}_{F,P}$ of rational functions on F at P is a discrete valuation ring with valuation given by:

$$\operatorname{ord}_P(h) := \dim_k \mathcal{O}_{F,P}/\langle h \rangle \qquad \text{for } h \in \mathcal{O}_{F,P}$$

Regarding $\varphi = f/g$ as an element of the fraction field of $\mathcal{O}_{F,P}$, we compute its valuation with the formula

$$\mu_P(f/g) := \operatorname{ord}_P(f) - \operatorname{ord}_P(g)$$

which we will call the *multiplicity* of φ at P. Now we define the divisor of φ as:

$$\operatorname{div} \varphi = \sum_{P \in F} \mu_P(\varphi) \cdot P$$

The divisor of a rational function essentially encodes the information of its zeros and poles. If φ has positive multiplicity at a point this means it has a zero of that degree at that point. Similarly, if φ has negative multiplicity at a point this means it has a pole of this degree at that point. Since the multiplicity at a point is a form of valuation, the following general results follow:

(a) The valuation is a group homomorphism to $\mathbb{Z}$. This means that $\mu_P(\varphi_1 \cdot \varphi_2) = \mu_P(\varphi_1) + \mu_P(\varphi_2)$ which implies $\operatorname{div}(\varphi_1 \cdot \varphi_2) = \operatorname{div} \varphi_1 + \operatorname{div} \varphi_2$.
(b) The ultrametric inequality, which states that $\mu_P(\varphi_1 + \varphi_2) \geq \min(\mu_P(\varphi_1), \mu_P(\varphi_2))$ with a guarantee of equality when $\mu_P(\varphi_1) \neq \mu_P(\varphi_2)$.

This concept should remind the reader of multiplicities of zeros and poles in complex analysis.

Remark 2.3. It is important to recognize that the definition of the multiplicity of some $f \in S_d(F)$ at a point can be rewritten as

$$\operatorname{ord}_P f = \dim_k \mathcal{O}_{F,P}/\langle f\rangle = \dim_k \frac{\mathcal{O}_{\mathbb{P}^2,P}/\langle F\rangle}{\langle f\rangle} = \dim_k \mathcal{O}_{\mathbb{P}^2,P}/\langle F, f\rangle$$

which is exactly the intersection number $\mu_P(F, f)$ of F and f at P. This allows us to restate some foundational results about intersection numbers in the language of divisors.

Theorem 2.4 (Bezout's Theorem with Divisors)**.** *Suppose that $f \in S(F)$ is homogeneous. Then:*

$$\deg \operatorname{div} f = \deg \sum_P \mu_P(f, F)P = \deg f \cdot \deg F$$

Proof. This is a reformulation of Bezout's theorem in the language of divisors which can be found in [3, §6, Rem. 6.27]. □

Theorem 2.5 (AF+BG Theorem with Divisors)**.** *Suppose that $h, g \in S(F)$ are two nonzero homogeneous polynomials and $\operatorname{div} h \geq \operatorname{div} g$. Then there exists a homogeneous $b \in S(F)$ of degree $\deg h - \deg g$ with $h = bg$ in $S(F)$.*

Proof. This proof is found in [3, §6, Prop. 6.28]. To recap, since F is nonsingular its local ring at any point P is a discrete valuation ring. Noether's condition at P is then equivalent to $\mu_P(h, F) \geq \mu_P(g, F)$. For this to be satisfied at every point is just the statement that $\operatorname{div} h \geq \operatorname{div} g$. □

Corollary 2.6. *If φ is a rational function then $\deg \operatorname{div} \varphi = 0$.*

Proof. Note that $\varphi = f/g$ for some $f, g \in S(F)$ of the same degree. Then

$$\deg \operatorname{div} \varphi = \deg \operatorname{div} f - \deg \operatorname{div} g = \deg f \cdot \deg F - \deg g \cdot \deg F = 0$$

by Theorem 2.4. □

Definition 2.7 (Picard Group)**.** The set of divisors originating as $\operatorname{div} \varphi$ for some rational function φ form a subgroup of $\operatorname{Div} F$ which we call $\operatorname{PDiv} F$ or the principal divisors of F. The quotient

$$\operatorname{Pic} F := \frac{\operatorname{Div} F}{\operatorname{PDiv} F}$$

is called the Picard group of F.

3. A Classical Result

We would now like to understand "how many" functions there are with prescribed zeros and poles. This is the essence of the Riemann-Roch theorem.

Definition 3.1 (Space of Functions)**.** Given a divisor D, we define

$$L(D) := \{\varphi \in K(F) : \operatorname{div} \varphi + D \geq 0\} \cup \{0\}$$

so that if D has some positive multiplicity at a point P, then $\varphi \in L(D)$ is allowed to have a pole of at most that order on P. If D has some negative multiplicity at a point P, then φ is required to have a zero of at least that order on P.

By the ultrametric inequality $L(D)$ is a vector space and we can define its dimension:

$$l(D) = \dim_k L(D)$$

Using this definition we now have a goal. That is to compute and understand $l(D)$ for general divisors. This is what the Riemann-Roch theorem will accomplish.

Remark 3.2. If two divisors D_1 and D_2 are equivalent in $\operatorname{Pic} F$ then they are related by a rational function $D_2 = D_1 + \operatorname{div}\phi$. In this case the map

$$\psi : L(D_1) \to L(D_2) \qquad \varphi \longmapsto \phi^{-1} \cdot \varphi$$

is an isomorphism with inverse $\varphi \mapsto \phi \cdot \varphi$.

Due to this the difference between divisors which are the same in the Picard group is less interesting. This is why we define the Picard group. We are quotienting the differences we don't care about to focus on what is interesting.

Example 3.3. If $\deg D < 0$ then $l(D) = 0$.

Proof. Suppose for the sake of contradiction that $\varphi \in L(D)$. By Corollary 2.6 $\deg \operatorname{div}\varphi = 0$ and

$$\deg(\operatorname{div}\varphi + D) = \deg D < 0$$

which contradicts the assumption that $\operatorname{div}\varphi + D \geq 0$. □

Lemma 3.4. *If D_1 and D_2 are two divisors with $D_1 \leq D_2$ then:*

$$l(D_1) \leq l(D_2) \leq l(D_1) + \deg(D_2 - D_1)$$

Proof. It is clear that $l(D_1) \leq l(D_2)$. Therefore we only need to show that $l(D_2) \leq l(D_1) + \deg(D_2 - D_1)$. Since D_2 is given by adding some number of points to D_1 it suffices by induction on $\deg(D_2 - D_1)$ to show that

$$l(D + P) \leq l(D) + 1$$

for any point $P \in F$. Choose a rational function t that is a uniformizer of the local ring at P, suppose that D has multiplicity n at P, and consider the map:

$$\psi : L(D + P) \to k \qquad \varphi \longmapsto (t^{n+1}\varphi)(P)$$

This map is well defined since

$$\mu_P(t^{n+1}\varphi) = n + 1 + \mu_P(\varphi) \geq 0$$

because $D+P$ has multiplicity at most $n+1$ at P. The kernel of the map is exactly those rational functions with $n + 1 + \mu_P(\varphi) \geq 1 \implies \mu_P(\varphi) \geq -n$, i.e. those in $L(D)$. Therefore we obtain one of the following exact sequences:

$$\begin{cases} 0 \to L(D) \underset{\text{incl}}{\to} L(D+P) \underset{\psi}{\to} 0, & \text{if } \operatorname{im}\psi \text{ is trivial} \\ 0 \to L(D) \underset{\text{incl}}{\to} L(D+P) \underset{\psi}{\to} k \to 0, & \text{if } \operatorname{im}\psi \text{ is nontrivial} \end{cases}$$

In the first case $l(D + P) = l(D) + 0 = l(D)$ and in the second $l(D + P) = l(D) + \dim k = l(D) + 1$. □

Theorem 3.5.

$$l(D) \leq \max(0, \deg D + 1)$$

Proof. If $\deg D < 0$ then we are done by Example 3.3. Otherwise pick a point P and consider $D' = D - (n+1)P$ where $n = \deg D$. Then by Example 3.3 $l(D') = 0$. Furthermore by Lemma 3.4

$$l(D) = l(D' + (n+1)P) \leq 0 + n + 1 \leq n + 1$$

as desired. □

In particular $l(0) = 1$, the one-dimensional space of constant functions. This result may be thought of as an algebraic analogue to the statement that any holomorphic (global regular) function on on a compact connected Riemann surface is constant. While this classical result is very interesting, it can be strengthened to an equality.

Theorem 3.6 (Riemann-Roch)**.** *For any line l not equal to F we claim that $K_F := (\deg F - 3) \operatorname{div} l$, called the canonical divisor, defines the same divisor class in the Picard group. Moreover if g is the genus of F then*

$$l(D) - l(K_F - D) = \deg D - g + 1$$

and $\deg K_F = 2g - 2$.

For now the reader should not worry about the exact definitions of K_F and g. In an informal sense K_F gives information about the structure of the differential forms and g gives information about the number of "handles" of F topologically.

While it is possible to prove this theorem in a very classical style, these proofs can sometimes feel computational, unmotivated, or miraculous. Therefore in order to give a more motivated proof we must introduce new machinery and move to a more general setting. Those who are interested in seeing a full classical proof are encouraged to read [3, §8, Cor. 8.17].

4. Line Bundles

In this section line bundles will be introduced. In particular, we will see that on a smooth proper curve a line bundle carries the information of a divisor. In this way we can prove theorems about divisors by understanding line bundles. The reader is expected to be comfortable with the basic language of schemes and sheaves.

Definition 4.1 (Sheaf of Modules)**.** A (pre)sheaf of modules on a scheme X is a (pre)sheaf which assigns to each open set $U \subseteq X$ an $\mathcal{O}_X(U)$-module $\mathscr{F}(U)$. If $U_2 \subseteq U_1$ then $\mathscr{F}$ is compatible with the restriction map $\phi : U_1 \to U_2$ in the sense that if $f \in \mathcal{O}_X(U_1)$ and $m \in \mathscr{F}(U_1)$ then:

$$\phi^*(f \cdot m) = \phi^*(f) \cdot \phi^*(m)$$

A morphism $\psi : \mathscr{F}_1 \to \mathscr{F}_2$ of (pre)sheaves of modules is a collection of $\mathcal{O}_X(U)$-module morphisms $\psi_U : \mathscr{F}_1(U) \to \mathscr{F}_2(U)$ which are compatible with the restriction maps.

Example 4.2. Suppose that $X = \operatorname{Spec} R$ is an affine scheme and M is an R-module. Then the sheaf $\widetilde{M}$ associated to M is defined as:

$$\begin{aligned}\widetilde{M}(U) := \{(\varphi_\mathfrak{p} \in M_\mathfrak{p})_{\mathfrak{p} \in U} : &\text{For every } \mathfrak{p} \in U \text{ there is an open neighborhood } U_\mathfrak{p} \text{ and} \\ &m \in M, f \in R \text{ with } \varphi_\mathfrak{q} = m/f \text{ for each } \mathfrak{q} \in U_\mathfrak{p}\}\end{aligned}$$

Therefore an element of $\widetilde{M}(U)$ is a collection of $\varphi_\mathfrak{p} \in M_\mathfrak{p}$ such that each point has an affine open neighborhood $U_\mathfrak{p}$ where $\varphi_\mathfrak{q}, \mathfrak{q} \in U_\mathfrak{p}$ are all represented algebraically by the same element of $R^{-1}M$. The motivation for this definition is if M, N are R-modules then maps $M \to N$ are in bijective correspondence with maps $\widetilde{M} \to \widetilde{N}$. In this way there is an equivalence of categories between R-modules and associated sheaves of $\operatorname{Spec} R$ [1, Ch. II, §5, Props. 5.1–5.5].

Definition 4.3 (Types of Sheaves of Modules)**.** Suppose that X is Noetherian and let $\mathscr{F}$ be a sheaf of modules on X. Then if for any point $\mathfrak{p} \in X$, we can always find a neighborhood U around $\mathfrak{p}$ satisfying the following hierarchical conditions, $\mathscr{F}$ may also be one of the following:

Quasi-coherent: There is an $\mathcal{O}_X(U)$-module M_U with $\mathscr{F}|_U \cong \widetilde{M_U}$.
Coherent: M_U is finitely generated over $\mathcal{O}_X(U)$.
Locally free: $M_U \cong \mathcal{O}_X(U)^{n_U}$ for some n_U.
Line bundle: $n_U = 1$ so that $M_U \cong \mathcal{O}_X(U)$.

By construction these are all local properties.

It is a theorem in [1, Ch. II, §5, Prop. 5.4] that quasi-coherence can be checked on any affine open cover. Furthermore if X is a Noetherian scheme then coherence can be checked on any affine open cover. In this paper we will be mainly concerned with line bundles.

For the remainder of the section C will denote an integral scheme of dimension 1, proper over k, all of whose local rings are regular. The sum of these quantifiers is to say that C is a proper smooth curve. The theory of these objects is developed in [1, Ch. IV, §1]. It is not strictly necessary for the reader to understand this, one may think of C as a curve as in the previous sections.

Example 4.4. Suppose that L is a line bundle over C, and let $\eta \in C$ be the generic point. Since C is irreducible, every nonempty open set contains η, and the nonempty open sets form a filtered system under reverse inclusion: for any two nonempty open sets U_1, U_2, the intersection $U_1 \cap U_2$ is a nonempty open set contained in both. Therefore we define the space of rational sections of L to be the stalk of L at the generic point

$$L_\eta := \varinjlim_{\eta \in U} L(U)$$

where the transition maps are the usual restriction maps. When $L = \mathcal{O}_C$, this stalk is

$$(\mathcal{O}_C)_\eta = K(C),$$

so an element can be identified with a function that is regular on some nonempty open subset, modulo agreement after restricting to a smaller nonempty open subset. For a general line bundle L, the space L_η is a one-dimensional $K(C)$-vector space. This is called the *space of rational sections of* L and will also be denoted

$$L \otimes_{\mathcal{O}_C} K(C).$$

Remark 4.5. Because C is a smooth proper curve there is a well defined notion of a divisor as a formal sum of points as in Definition 2.1 [1, Ch. II, §6]. Furthermore given a rational function $\varphi \in K(C)$ there is a well defined notion of the divisor of φ as in Construction 2.2 [1, Ch. II, §6].

Proof. A complete treatment of divisors of curves is found in [1, Ch. II, §6]. Essentially, an element $\varphi \in K(C)$ can be considered as an element of the fraction field of the local ring $\mathcal{O}_{C,P}$ at a point. Since the local ring is a discrete valuation ring by assumption we can take the multiplicity at P to be the valuation $\mu_P(\varphi)$. □

Construction 4.6 (Line Bundle Associated to a Divisor)**.** For a divisor D we can form the *associated line bundle* as

$$\mathcal{L}^D(U) = \{f \in K(C) : \operatorname{div} f|_U + D|_U \geq 0\}$$

where the notation $D|_U$ means that we are ignoring all points outside of U.

Proof. We need to show that $\mathcal{L}^D$ is a line bundle. Since C is a smooth curve all its points (excluding the generic point) are closed and it is a T_1 space with respect to these points. This will be used implicitly in the proof.

Suppose that $P \in C$. Let U be a neighborhood of P which does not contain any other point on which D has nonzero multiplicity. If D has 0 multiplicity on P then we are done by definition. Otherwise let $n = \mu_P(D)$. Since C is a smooth curve the local ring $\mathcal{O}_{C,P}$ is a discrete valuation ring so let $t \in K(C)$ be a uniformizing parameter. We can choose another open neighborhood $U' \subseteq U$ of P where t is regular and has no other zeros or poles in U'. In this neighborhood

$$\mathcal{L}^D(U') = \{f \in K(C) : \operatorname{div} f|_{U'} + nP \geq 0\}$$

and we can define the maps

$$\psi : \mathcal{L}^D(U') \to \mathcal{O}_C(U'), \varphi \mapsto t^n\varphi \qquad \psi : \mathcal{O}_C(U') \to \mathcal{L}^D(U'), \varphi \mapsto t^{-n}\varphi$$

which are mutually inverse and therefore define an isomorphism $\mathcal{L}^D|_{U'} \cong \mathcal{O}_C|_{U'}$. □

Intuitively $\mathcal{L}^D$ carries the information of the functions satisfying the divisor D locally. One of the overarching themes in algebraic geometry is that forcing things to be local makes them easier to work with. This is another example of this principle.

Construction 4.7 (Divisor of Line Bundle)**.** Suppose that L is a line bundle over C and $s \in L \otimes_{\mathcal{O}_C} K(C)$ is a nonzero rational section. At any point $P \in C$ we may choose an affine open U where there is an isomorphism $L|_U \cong \mathcal{O}_C|_U$. This identifies s with a rational function $\varphi \in K(C)$ locally around P. Therefore define $\mu_P(s) := \mu_P(\varphi)$. Since we can do this around any point we may define:

$$\operatorname{div} s := \sum_{P \in C} \mu_P(s) \cdot P$$

In order for this to be well defined, we need to prove that our choice of isomorphism $L|_U \cong \mathcal{O}_C|_U$ does not matter. In particular if $\psi_1 : L|_U \to \mathcal{O}_C|_U$ and $\psi_2 : L|_U \to \mathcal{O}_C|_U$ are two isomorphisms and $\varphi_i = \psi_i(s)$ then

$$\varphi_2 = \psi_2(s) = \psi_2\psi_1^{-1}\psi_1(s) = \psi_2\psi_1^{-1}(\varphi_1)$$

however $\psi_2\psi_1^{-1}$ is an isomorphism $\mathcal{O}_C|_U \to \mathcal{O}_C|_U$. Therefore it is given by multiplication by a regular function $\phi \in \mathcal{O}_C|_U$ that is nowhere 0 on U. In particular on U

$$\varphi_2 = \phi\varphi_1 \implies \operatorname{div}\varphi_2 = \operatorname{div}\phi + \operatorname{div}\varphi_1 \implies \mu_P(\varphi_1) = \mu_P(\varphi_2)$$

since ϕ does not have multiplicity at P by construction.

We would like to define the divisor of a line bundle as the divisor of one of its rational sections. Unfortunately, this is noncanonical since we are arbitrarily choosing a rational section. However since $L \otimes_{\mathcal{O}_C} K(C)$ is a 1-dimensional $K(C)$ vector space our choice is actually well defined up to multiplication by a rational function. This motivates the following theorem.

Theorem 4.8. *If we define* $\operatorname{Pic}' C$ *as the group of isomorphism classes of line bundles over* C *under the tensor product* $\otimes_{\mathcal{O}_C}$ *then the maps*

$$\psi : \operatorname{Pic} C \to \operatorname{Pic}' C, D \mapsto \mathcal{L}^D \qquad \psi^{-1} : \operatorname{Pic}' C \to \operatorname{Pic} C, L \mapsto \operatorname{div} s$$

given in Construction 4.6 and Construction 4.7 are mutually inverse isomorphisms of abelian groups.

Proof. There are a few things to check:

(a) The choice of s does not change the image in the Picard group.
(b) L is isomorphic to $\mathcal{L}^{\operatorname{div} s}$.
(c) Two equivalent divisors D_1 and D_2 define the same line bundle.
(d) $\mathcal{L}^{D_1} \otimes_{\mathcal{O}_C} \mathcal{L}^{D_2} \cong \mathcal{L}^{D_1+D_2}$.

Our proofs will all rely heavily on the structure of $\mathcal{L} \otimes_{\mathcal{O}_C} K(C)$ as a 1-dimensional $K(C)$ vector space.

(a) Note that s is a basis for the space of rational sections of L, any other rational section can be written uniquely as st for $t \in K(C)$. If we choose another section $s' = st$, then $s'/s = t$ is a rational function and $\operatorname{div} t = \operatorname{div} s' - \operatorname{div} s$ is trivial in the Picard group.
(b) As before note every rational function is of the form st and consider the map:
$$\mathcal{L} \otimes_{\mathcal{O}_C} K(C) \to K(C) \qquad st \longmapsto t$$
The condition that st is regular on some open set U is that $\operatorname{div} s|_U + \operatorname{div} t|_U \geq 0$. This is exactly the condition that $t \in \mathcal{L}^{\operatorname{div} s}(U)$ so the map defines an isomorphism $L \cong \mathcal{L}^{\operatorname{div} s}$.
(c) Suppose that $D_1 - D_2 = \operatorname{div} \phi$; then the map
$$\mathcal{L}^{D_1} \otimes K(C) \to \mathcal{L}^{D_2} \otimes K(C) \qquad \varphi \mapsto \varphi \cdot \phi$$
gives an isomorphism $\mathcal{L}^{D_1} \cong \mathcal{L}^{D_2}$.
(d) The rational sections of $\mathcal{L}^{D_1} \otimes_{\mathcal{O}_C} \mathcal{L}^{D_2}$ is a tensor product of 1-dimensional $K(C)$ vector spaces. Therefore for any $a \in \mathcal{L}^{D_1}$ and $b \in \mathcal{L}^{D_2}$ every rational function is of the form tab for some $t \in K(C)$. Furthermore we can assume that $\operatorname{div} a = D_1$ and $\operatorname{div} b = D_2$ by (b). The condition that tab is regular on some open set U is that $\operatorname{div} t|_U + \operatorname{div} a|_U + \operatorname{div} b|_U \geq 0$. This is just the condition that $t \in \mathcal{L}^{D_1+D_2}(U)$ as desired.

□

In this way, the line bundles encode the exact same information as the Picard group of divisors. This makes them an extremely useful bookkeeping device for reasoning about divisors. We define the degree of a line bundle using the degree of the associated divisor:
$$\deg L := \deg D \qquad \text{when } L \cong \mathcal{L}^D$$

Construction 4.9 (Dual Bundle)**.** Given a line bundle L over C we define the *dual bundle* as
$$L^\vee := \operatorname{Hom}_{\mathcal{O}_C}(L, \mathcal{O}_C)$$
and claim that $L \otimes L^\vee \cong \mathcal{O}_C$. In particular $(\mathcal{L}^D)^\vee \cong \mathcal{L}^{-D}$.

Proof. On any open U where L is trivial the evaluation morphism $\mathrm{ev} : L \otimes L^\vee \to \mathcal{O}_C$ is an isomorphism. This is because we can (noncanonically) identify $L|_U$ with $\mathcal{O}_C|_U$ and write:
$$\mathcal{O}_C|_U \otimes \operatorname{Hom}_{\mathcal{O}_C|U}(\mathcal{O}_C|_U, \mathcal{O}_C|_U) \cong \mathcal{O}_C|_U \otimes \mathcal{O}_C|_U \cong \mathcal{O}_C|_U$$
Since ev is a canonical isomorphism it extends to a global isomorphism as desired.

□

For this reason line bundles are often called *invertible sheaves* since they have "inverses" in the sense of the tensor product.

Theorem 4.10. *Suppose that L is a line bundle over C and D is an effective divisor. Then there is an exact sequence*

$$0 \to L \to L \otimes_{\mathcal{O}_C} \mathcal{L}^D \to Q^D \to 0$$

where Q^D is a skyscraper sheaf supported on $\operatorname{Supp}(D)$ *with length* $\deg D$.

Proof. Firstly suppose that $L = \mathcal{L}^{D'}$ for some divisor D'. Since D is effective there is the obvious inclusion $\mathcal{L}^{D'} \to \mathcal{L}^{D'+D}$. We need to prove that the quotient sheaf of this inclusion satisfies the stated properties. Localizing the sequence at a point P yields:

$$0 \to \mathcal{L}_P^{D'} \to \mathcal{L}_P^{D'+D} \to Q_P^D \to 0$$

We choose a local uniformizer π at P and denote $\mu_P(D') = a$ and $\mu_P(D) = b$. Then we have the identifications

$$\mathcal{L}_P^{D'} = \pi^{-a}\mathcal{O}_{C,P} \qquad \mathcal{L}_P^{D'+D} = \pi^{-a-b}\mathcal{O}_{C,P}$$

and the quotient of the inclusion is

$$Q_P^D = \frac{\pi^{-a-b}\mathcal{O}_{C,P}}{\pi^{-a}\mathcal{O}_{C,P}}$$

which is isomorphic to $\mathcal{O}_{C,P}/(\pi^b)$ as an $\mathcal{O}_{C,P}$-module. When $b = 0$ this is 0, so Q^D is indeed supported only on $\operatorname{Supp}(D)$. Moreover this is a b-dimensional k-vector space, so the length of Q_D is

$$\sum_{P \in C} \mu_p(D) = \deg D$$

as desired. □

Example 4.11. One of the simplest examples of line bundles is those over $\mathbb{P}^1$. Note that over $\mathbb{P}^1$ the rational function defined by a Möbius transformation can be chosen to have a pole and a zero of order 1 on any 2 points we choose. Therefore when we quotient by the rational divisors in the Picard group we lose all the information of where the poles and zeros of a rational function are. Due to this line bundles over $\mathbb{P}^1$ are characterized by their degree and we define the *twisting sheaf* as

$$\mathcal{O}_{\mathbb{P}^1}(n) := \mathcal{L}^{nP}$$

for any point P.

5. Kähler Differentials and the Canonical Divisor

In this section the canonical divisor will be defined and motivated using the machinery of Kähler differentials. For the remainder of this section let R be a k-algebra, M an R-module, and S a manifold.

Definition 5.1 (Derivations)**.** A map $\delta : R \to M$ is called a derivation if it satisfies the conditions:

Additivity: $\delta(f + g) = \delta(f) + \delta(g)$ for all $f, g \in R$.
Leibniz Rule: $\delta(fg) = f\delta(g) + g\delta(f)$ for all $f, g \in R$.
k-linearity: $\delta(sf) = s\delta(f)$ for all $f \in R, s \in k$.

The space of derivations is denoted $\mathrm{Der}_{R/k}(M)$. Note that there is an action of R on this space by
$$(r \cdot \delta)(a) = r \cdot \delta(a)$$
making $\mathrm{Der}_{R/k}(M)$ an R-module.

In algebraic geometry rings should be thought of as the functions on a space. In the context of manifolds, we think of R as the smooth functions on a manifold. Then one learns that a derivation at a point p is an operator which satisfies the Leibniz rule
$$\delta(fg) = f(p)\delta(g) + g(p)\delta(f)$$
and the set of all derivations is the tangent space T_pS [9, Ch. 1, §2.3]. Thinking algebraically, take $M = R/\mathfrak{m} \cong k$ for some maximal ideal $\mathfrak{m}$. Here, $\mathfrak{m}$ corresponds to the ideal of smooth functions vanishing at a point p on the manifold and the natural map $R \to M$ is evaluation at p. Therefore the derivations of the map $R \to M$ are by definition the tangent space of the manifold at p. In the algebraic context, we take this as the definition and derive a lot of familiar results. Essentially, the map $R \to M$ corresponds to evaluation of some $f \in R$ at the closed point $\mathfrak{m} \in \mathrm{Spec}(R)$. In this way $\mathrm{Der}_{R/k}(M)$ corresponds to the tangent space of $\mathrm{Spec}(R)$ at $\mathfrak{m}$.

Remark 5.2.
If $A \subseteq R$ then the condition that some $\delta \in \mathrm{Der}_{R/k}(M)$ is A-linear is equivalent to $A \subseteq \ker(\delta)$.

Proof. Suppose that $a \in A$.
⇒: Note $\delta(1) = \delta(1 \cdot 1) = \delta(1) + \delta(1)$ so $\delta(1) = 0$. Now $\delta(a) = a\delta(1) = 0$.
⇐: Note $\delta(ar) = a\delta(r) + r\delta(a) = a\delta(r)$.

□

Therefore instead of requiring that δ be k-linear we could have required that δ kills k.

Definition 5.3 (Kähler Differentials)**.** We define $\Omega_{R/k}$ to be the free R module generated by the formal symbols df modulo the relations

- $d(f + g) = df + dg$ for all $f, g \in R$;
- $d(fg) = fdg + gdf$ for all $f, g \in R$;
- $ds = 0$ for all $s \in k$.

We equip $\Omega_{R/k}$ with a map $d : R \to \Omega_{R/k}, f \mapsto df$ called the *universal derivation*. One can check that this map is a derivation.

In the context of manifolds, one can think of $\Omega_{R/k}$ as the space of 1-forms. The universal derivation is analogous to the exterior derivative, which maps 0-forms, or functions, onto 1-forms. This idea will become much clearer with the following theorem.

Theorem 5.4 (Universal Property of $\Omega_{R/k}$)**.** *Given a derivation $\delta : R \to M$ there is a unique R-module map $\phi : \Omega_{R/k} \to M$ that makes the diagram commute*

$$\begin{tikzcd} R \arrow[r, "\delta"] \arrow[d, "d"] & M \\ \Omega_{R/k} \arrow[ur, "\phi"] & \end{tikzcd}$$

so that $\mathrm{Der}_{R/k}(M) = \mathrm{Hom}_R(\Omega_{R/k}, M)$*.*

Proof. Suppose that $f, g \in R$ and $s \in k$. Commutativity of the diagram requires that $\phi(df) = \delta(f)$ which determines the map uniquely. We need to show that all the relations in Definition 5.3 are satisfied.

- $\phi(d(f+g)) = \delta(f+g) = \delta(f) + \delta(g) = \phi(df) + \phi(dg)$;
- $\phi(d(fg)) = \delta(fg) = f\delta(g) + g\delta(f) = \phi(fdg + gdf)$;
- $\phi(ds) = \delta(s) = 0$.

□

We can express this result in the language of manifolds. Recall that when $M = R/\mathfrak{m} \cong k$ the derivations $\mathrm{Der}_{A/k}$ correspond to the tangent space at $\mathfrak{m}$. In manifolds, the first order differential forms are global sections of the dual space of the tangent space. This means that at a point, an element of the tangent space $x \in T_pS$ corresponds to the map $\omega \mapsto \omega(x)$ which evaluates a differential form on x. This gives a correspondence between T_pS and maps $T_p^*S \to k$. The equation $\mathrm{Der}_{R/k}(M) = \mathrm{Hom}(\Omega_{R/k}, M)$ expresses this fact very directly in the algebraic context.

Kähler differentials are very difficult to compute by hand in general, however there exist two powerful exact sequences which allow one to compute Kähler differentials for most "reasonable" rings.

Lemma 5.5 (First Exact Sequence)**.** *If B is an R-algebra then there is the exact sequence*

$$\Omega_{R/k} \otimes_R B \to \Omega_{B/k} \to \Omega_{B/R} \to 0$$

of B-modules.

Proof. It suffices to show that the sequence

$$0 \to \mathrm{Hom}_B(\Omega_{B/R}, M) \to \mathrm{Hom}_B(\Omega_{B/k}, M) \to \mathrm{Hom}_B(\Omega_{R/k} \otimes_R B, M)$$

is exact for every B-module M. Note that

$$\mathrm{Hom}_B(\Omega_{R/k} \otimes_R B, M) \cong \mathrm{Hom}_R(\Omega_{R/k}, M)$$

by adjointness of extension/restriction of scalars. Then by Theorem 5.4 this sequence corresponds to:

$$0 \to \mathrm{Der}_{B/R}(M) \to \mathrm{Der}_{B/k}(M) \to \mathrm{Der}_{R/k}(M)$$

The map $\mathrm{Der}_{B/k}(M) \to \mathrm{Der}_{R/k}(M)$ is the pullback to R. The kernel are those maps which kill R, which is equivalent to being R-linear by Remark 5.2. Therefore the kernel is exactly $\mathrm{Der}_{B/R}(M)$ as desired. □

Geometrically, this sequence relates the differential forms on the domain and codomain of a map. The algebra homomorphism $R \to B$ corresponds to the morphism $f : Y \to X$ where $Y = \operatorname{Spec} B$ and $X = \operatorname{Spec} R$. Then the terms in the exact sequence correspond to the following geometric objects:

- $\Omega_{R/k} \otimes_R B$ corresponds to the *pullback* $f^*\Omega_X$, representing forms on X transported to Y.
- $\Omega_{B/k}$ corresponds to Ω_Y, the intrinsic differential forms on Y.
- $\Omega_{B/R}$ corresponds to the *relative differentials* $\Omega_{Y/X}$.

The first map is the pullback of forms. The relative differentials $\Omega_{B/R}$ capture the "vertical" geometry of the map; they represent the differential forms along the fibers of f by measuring the infinitesimal changes in Y that vanish in the directions coming from X.

Lemma 5.6 (Second Exact Sequence)**.** *If I is an ideal of R and $B = R/I$ then there is the exact sequence*

$$I/I^2 \to \Omega_{R/k} \otimes_R B \to \Omega_{B/k} \to 0$$

of B-modules.

Proof. It suffices to show that the sequence

$$0 \to \mathrm{Hom}_B(\Omega_{B/k}, M) \to \mathrm{Hom}_B(\Omega_{R/k} \otimes_R B, M) \to \mathrm{Hom}_B(I/I^2, M)$$

is exact for every B-module M. Note that

$$\mathrm{Hom}_B(\Omega_{R/k} \otimes_R B, M) \cong \mathrm{Hom}_R(\Omega_{R/k}, M) \qquad \mathrm{Hom}_B(I/I^2, M) \cong \mathrm{Hom}_R(I, M)$$

by adjointness of extension/restriction of scalars and the universal property of quotients respectively. Now we have

$$0 \to \mathrm{Hom}_B(\Omega_{B/k}, M) \to \mathrm{Hom}_R(\Omega_{R/k}, M) \to \mathrm{Hom}_R(I, M)$$

which by Theorem 5.4 corresponds to:

$$0 \to \mathrm{Der}_{B/k}(M) \to \mathrm{Der}_{R/k}(M) \to \mathrm{Hom}_R(I, M)$$

Then for $i \in I$ and $f \in R$

$$\delta(if) = i\delta(f) + f\delta(i) = f\delta(i)$$

since I kills M (it is an R/I-module). Therefore the map $\mathrm{Der}_{R/k}(M) \to \mathrm{Hom}_R(I, M)$ by restriction to I is well defined. By comparing definitions we see that an element of $\mathrm{Der}_{B/k}$ is exactly an element of $\mathrm{Der}_{R/k}$ (via precomposition with the quotient map) that kills I. This is exactly the kernel of the map $\mathrm{Der}_{R/k}(M) \to \mathrm{Hom}_R(I, M)$ as desired. □

Geometrically, this sequence relates the cotangent bundle of a submanifold $Y = \mathrm{Spec}\, B$ to its ambient space $X = \mathrm{Spec}\, R$. Then the terms in the exact sequence correspond to the following geometric objects:

- $\Omega_{R/k} \otimes_R B$ corresponds to $T^*X|_Y$, the cotangent bundle of the ambient space restricted to the submanifold.
- $\Omega_{B/k}$ corresponds to T^*Y, the intrinsic cotangent bundle of the submanifold.
- I/I^2 corresponds to the *conormal bundle* N^*Y.

The sequence states that the differential forms on Y are obtained by taking the ambient differential forms restricted to Y and ”modding out” by the conormal directions (the differentials of the constraint equations defining Y). In the smooth manifold context, this map is injective on the left, reflecting the decomposition of the ambient cotangent space into tangent and normal components.

Example 5.7 (Cotangent Space)**.** Suppose that $\mathfrak{m}$ is the kernel of a map $R \to k$. By applying Lemma 5.6 to $R/\mathfrak{m}$ we obtain the exact sequence

$$\mathfrak{m}/\mathfrak{m}^2 \to \Omega_{R/k} \otimes_R (R/\mathfrak{m}) \to 0$$

and see that $\mathfrak{m}/\mathfrak{m}^2$ surjects onto $\Omega_{R/k} \otimes_R k$.

Example 5.7 says that $\mathfrak{m}/\mathfrak{m}^2$ surjects onto the cotangent space of R at $\mathfrak{m}$. As k-vector spaces we can write $R \cong \mathfrak{m} \oplus k$. A derivation of the map $R \to k$ sends k to 0 and is determined by its restriction to $\mathfrak{m}$. However it must also send $\mathfrak{m}^2$ to 0 by the Leibniz rule. Therefore the derivations are in correspondence with k-linear forms on $\mathfrak{m}/\mathfrak{m}^2$. Then $\mathfrak{m}/\mathfrak{m}^2$ is dual to the tangent space at $\mathfrak{m}$, proving the stronger result that $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{R/k} \otimes_R k$.

This is important for the following reason. We would like to say that $\operatorname{Spec} R$ is *smooth* if the tangent space at every point has the same dimension as the Krull dimension of R. Since $\Omega_{R/k}$ is locally dual to the tangent space we can consider instead when it has this property. Indeed, Example 5.7 shows that R must be a *regular ring* in order for $\operatorname{Spec} R$ to be smooth. Through some work it is possible to show that if R is a k-algebra of finite type then $\operatorname{Spec} R$ is smooth if and only if R is a regular ring. This gives a nice algebraic characterization of this property.

Theorem 5.8. *Suppose that R is a finitely generated smooth k-algebra of pure dimension 1. Then, for every prime ideal $\mathfrak{p} \subseteq R$, the localized module*

$$\Omega_{R/k,\mathfrak{p}}$$

is a free $R_{\mathfrak{p}}$-module of rank 1.

Proof. Since the claim is local, after shrinking around $\mathfrak{p}$ and reordering the coordinates, we may write

$$R = k[x_1, \dots, x_n]/(f_1, \dots, f_{n-1})$$

so that

$$A = \left(\frac{\partial f_i}{\partial x_j}\right)_{1 \le i,j \le n-1}$$

is invertible over $R_{\mathfrak{p}}$ by the Jacobian criterion [1, Ch. I, §5, Thm. 5.1]. The module $\Omega_{R/k}$ is generated by $dx_1, \dots, dx_n$, subject to the relations:

$$df_i = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}\, dx_j = 0, \qquad 1 \le i \le n-1$$

After localizing at $\mathfrak{p}$, set:

$$A = \left(\frac{\partial f_i}{\partial x_j}\right)_{1 \le i,j \le n-1} \qquad b = \left(\frac{\partial f_i}{\partial x_n}\right)_{1 \le i \le n-1}$$

Thus the relations $df_i = 0$ give

$$\begin{pmatrix} dx_1 \\ \vdots \\ dx_{n-1} \end{pmatrix} = -A^{-1} b\, dx_n$$

and $\Omega_{R/k,\mathfrak{p}}$ is generated by dx_n.

To see that there is no relation on dx_n, write $A^{-1}b = (c_1, \dots, c_{n-1})^T$. The assignment

$$dx_n \longmapsto 1 \qquad dx_j \longmapsto -c_j \quad (1 \le j \le n-1)$$

respects all the relations $df_i = 0$, so it defines an $R_{\mathfrak{p}}$-linear map

$$\Omega_{R/k,\mathfrak{p}} \longrightarrow R_{\mathfrak{p}}$$

sending dx_n to 1. Therefore dx_n is a basis, and:

$$\Omega_{R/k,\mathfrak{p}} \cong R_{\mathfrak{p}}$$

□

Remark 5.9. Suppose that $\operatorname{Spec} R$ is a regular one dimensional smooth affine variety. Pick any nonzero differential $\omega \in \Omega_{R/k}$. At a point $\mathfrak{p} \in \operatorname{Spec}(R)$ the local ring $R_\mathfrak{p}$ is a discrete valuation ring so we can choose a uniformizer t. Then by Theorem 5.8 the stalk ω can be written uniquely as fdt for some $f \in R_\mathfrak{p}$, so we define $\mu_\mathfrak{p}(\omega) = \mu_\mathfrak{p}(f)$. Since we can define the multiplicity of ω locally around every point, we can define its divisor using this notion. It turns out that any differential form $\omega \in \Omega_{R/k}$ gives the same divisor class in the Picard group. This divisor class is therefore called the *canonical divisor*. An algebraic proof of this statement is hard, so we will instead approach it via the machinery of line bundles.

In the language of manifolds, a top degree differential form ω be locally associated to a function f by choosing a basis $x_1, \ldots, x_n$ of the coordinate chart and writing:

$$\omega = f dx_1 \wedge \cdots \wedge dx_n$$

We can use this local notion to associate ω to a well defined divisor, which turns out not to depend on the choice of coordinates. Moreover under reasonable assumptions every differential form defines the same divisor class in the Picard group.

As the reader may have guessed, the "better" definition of the canonical divisor is to use $\Omega_{R/k}$ to define a line bundle. Then we can take the divisor associated to the line bundle. If we trace back definitions this is actually equivalent to our naive definition, however the machinery of line bundles allow us to avoid algebraic technicalities.

Lemma 5.10. *Consider the map*

$$\delta : R \otimes_k R \to R \qquad f \otimes g \longmapsto fg$$

and set $J := \ker \delta$. *Then* $J/J^2 \cong \Omega_{R/k}$.

Proof. Consider $R \otimes_k R$ as an R-module via left multiplication. There are the isomorphisms

$$J/J^2 \to \Omega_R \qquad f_i \otimes g_i \longmapsto f_i dg_i$$
$$\Omega_R \to J/J^2 \qquad df \mapsto 1 \otimes f - f \otimes 1$$

which are inverses of each other. The details are worked out in [4, §15.4]. □

Lemma 5.11 (Ideal Sheaf)**.** *Suppose that X is a separated Noetherian scheme and $Y \hookrightarrow X$ is a closed subscheme. The inclusion $i : Y \to X$ induces the following constructions:*

- *The pushforward sheaf* $(i_*\mathcal{O}_Y)(U) = \mathcal{O}_Y(i^{-1}(U))$ *on* X.
- *The morphism of sheaves*

$$i^\sharp : \mathcal{O}_X \to i_*\mathcal{O}_Y \qquad \varphi \longmapsto i^*\varphi$$

  induced by the pullbacks $i_U^* : \mathcal{O}_X(U) \to \mathcal{O}_Y(i^{-1}(U))$.
- *The ideal sheaf* $\mathscr{I}_{Y/X}(U) = \ker i_U^*$ *which is the kernel of* $i^\sharp$.
- *The exact sequence* $0 \to \mathscr{I}_{Y/X} \to \mathcal{O}_X \to i_*\mathcal{O}_Y \to 0$.

Proof. Since all the constructions used preserve the sheaf property [4, §13] we can check exactness locally. In particular we can assume that $X = \operatorname{Spec} R$ is affine. In this case $\mathcal{O}_X$ and $i_*\mathcal{O}_Y$ are the sheaves associated to the R-modules R and R/J, where $J := \mathcal{I}(Y)$. Therefore this sequence corresponds to the exact sequence:

$$0 \to J \to R \to R/J \to 0$$

□

Construction 5.12 (Cotangent Sheaf)**.** Suppose that X is a separated k-scheme. Then the map

$$i : X \to X \times X \qquad x \longmapsto x \times x$$

is a closed immersion which identifies X with the diagonal $\Delta_X \subseteq X \times_k X$. On an affine open $U = \operatorname{Spec}(R) \subseteq X$ the pullback map $\mathcal{O}_{X \times_k X} \to i_* \mathcal{O}_X$ corresponds exactly to

$$\delta : R \otimes_k R \to R \qquad a \otimes b \longmapsto ab$$

therefore by Lemma 5.11 the ideal sheaf $\mathscr{I}$ induced by i is identified with $\ker \delta = J$ so:

$$\mathscr{I}|_{U \times_k U} \cong \widetilde{J}.$$

Now take

$$\Omega_X = i^*(\mathscr{I}/\mathscr{I}^2)$$

to be the pullback sheaf of $\mathscr{I}/\mathscr{I}^2$. On an affine open the pullback just considers J/J^2 as an R-module instead of an $R \otimes_k R$-module [4, §15.5]. Therefore by Lemma 5.10

$$\Omega_X|_U \cong \widetilde{J/J^2} \cong \widetilde{\Omega_{R/k}}$$

on some affine open subset $U \cong \operatorname{Spec} R$.

We have successfully constructed a sheaf which is identified with $\Omega_{R/k}$ on affine opens. To prove it is a line bundle, we will use an algebraic argument to prove a result which validates our intuition about differential forms.

Proposition 5.13. *If C is a smooth curve, then Ω_C is locally free of rank 1.*

Proof. By Theorem 5.8, every stalk $\Omega_{C,P}$ is a free $\mathcal{O}_{C,P}$-module of rank 1. Moreover, Ω_C is coherent, since on every affine open $U = \operatorname{Spec} R$ the module $\Omega_{R/k}$ is finitely generated and R is Noetherian. Therefore Ω_C is locally free of rank 1 [1, Ch. II, §5, Ex. 5.7(a)–(b)]. □

Now that we have Proposition 5.13 we can finally give a rigorous definition of the canonical divisor.

Construction 5.14. (Canonical Bundle) Note that if C is a smooth proper curve then by Proposition 5.13 Ω_C is a line bundle called the *canonical bundle*, and hence defines a unique divisor class in the Picard group called the *canonical divisor*.

The reader is encouraged to try to unravel the definitions and see why this definition of the canonical divisor is the same as the one given in Remark 5.9.

Theorem 5.15. *On $\mathbb{P}^1$ we have*

$$\Omega_{\mathbb{P}^1} \simeq \mathcal{O}_{\mathbb{P}^1}(-2).$$

Proof. Let $U_0 = \operatorname{Spec} k[x]$ and $U_\infty = \operatorname{Spec} k[u]$, with $u = x^{-1}$ on $U_0 \cap U_\infty$. On U_0, the sheaf $\Omega_{\mathbb{P}^1}$ is generated by dx, and on U_∞ it is generated by du. On the overlap,

$$du = d(x^{-1}) = -x^{-2}dx.$$

Replacing the local frame du by $-du$, the transition function becomes x^{-2}. On the other hand, $\mathcal{O}_{\mathbb{P}^1}(-2) = L^{-2\infty}$ has local frame 1 on U_0 and local frame u^2 on U_∞, and on the overlap $u^2 = x^{-2}$. Therefore $\Omega_{\mathbb{P}^1}$ and $\mathcal{O}_{\mathbb{P}^1}(-2)$ have the same transition function, so they are isomorphic. □

6. Cohomology of Sheaves and Cheap Riemann-Roch

In this section the tool of Čech Cohomology will be developed and motivated. This will prove extremely useful in giving a sophisticated and systematic proof of the Riemann-Roch theorem.

Definition 6.1 (Čech Complex)**.** Suppose that $U_1, \ldots, U_r$ is an affine open cover of X and $\mathscr{F}$ is a sheaf on X. The *Čech complex* with respect to this cover is

$$C^p(\mathscr{F}) := \bigoplus_{i_0<\cdots<i_p} \mathscr{F}(U_{i_0} \cap \cdots \cap U_{i_p})|_{U_{i_0}\cap\cdots\cap U_{i_p}}$$

so that an element $\varphi \in C^p(\mathscr{F})$ is a collection of sections $\varphi_{i_0<\cdots<i_p} \in \mathscr{F}(U_{i_0} \cap \cdots \cap U_{i_p})$ on each intersection of $p+1$ open sets in the cover. We impose no restriction on the sections themselves. There is a *boundary map* $d_p : C^p(\mathscr{F}) \to C^{p+1}(\mathscr{F})$ defined by:

$$(d_p\varphi)_{i_0,\ldots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \varphi_{i_0,\ldots,\hat{i}_k,\ldots,i_{p+1}}|_{U_{i_0}\cap\cdots\cap U_{i_{p+1}}}$$

Definition 6.2 (Čech Cohomology)**.** It can be verified via a combinatorial argument that $d^{p+1} \circ d^p = 0$ [1, Ch. III, §4]. Therefore we form the *cochain complex*

$$C^0(\mathscr{F}) \xrightarrow{d^0} C^1(\mathscr{F}) \xrightarrow{d^1} C^2(\mathscr{F}) \xrightarrow{d^2} \cdots$$

and define

$$H^p(X, \mathscr{F}) := \frac{\ker d^p}{\operatorname{im} d^{p-1}}$$

where $H^p(X, \mathscr{F})$ is the *p-th cohomology* of $\mathscr{F}$.

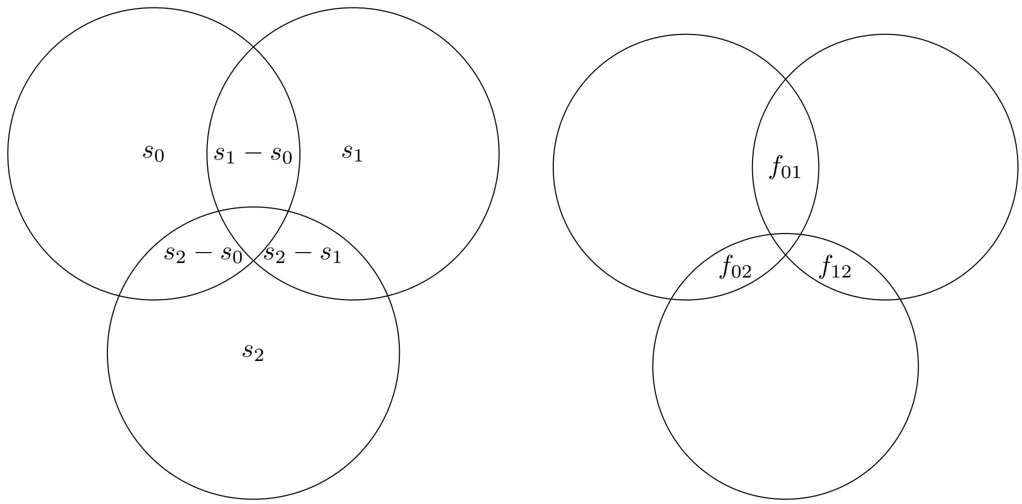
The Čech Cohomology essentially measures "how much" a collection of global sections can disagree. To see this we will discuss H^0 and H^1. Firstly $H^0(X, \mathscr{F})$ is just the kernel of d^0. Therefore write

$$(d^0 f)_{i_0,i_1} = f_{i_1}|_{U_{i_0}\cap U_{i_1}} - f_{i_0}|_{U_{i_0}\cap U_{i_1}} = 0 \implies f_{i_0}|_{U_{i_0}\cap U_{i_1}} = f_{i_1}|_{U_{i_0}\cap U_{i_1}}$$

which is exactly the sheaf condition that the $(f)_{i_1,i_2}$ are compatible on overlaps. In this way $H^0(X, \mathscr{F})$ corresponds to the *global sections* of $\mathscr{F}$, or those local sections which agree to "0th order". Now consider $H^1(X, \mathscr{F})$. A cocycle of $H^1(X, \mathscr{F})$ satisfies

$$(d^1 f)_{i_0,i_1,i_2} = (f_{i_1,i_2} - f_{i_0,i_2} + f_{i_0,i_1})|_{U_{i_0}\cap U_{i_1}\cap U_{i_2}} = 0$$

and a coboundary of $H^1(X, \mathscr{F})$ is the image of something in $C^0(X, \mathscr{F})$. This can be visualized as a Venn diagram:



We can consider the $s_i - s_j$ to be the "disagreements" of the sections s_i on the overlaps. The statement $d^1 \circ d^0(s_i) = 0$ says that the disagreement of s_0 and s_2 is the sum of the disagreements between s_0 and s_1, and s_1 and s_2. This can be viewed as a "first order" agreement condition. The cocycle condition means that we pick some f_{ij} on the intersections with this agreement property. However, we are not interested in f_{ij} which arise from some sections s_i. Therefore the homology asks, if the f_{ij} satisfy the first order agreement property, how might we be unable to glue them together into a system of sections s_i? In the same way $H^p(X, \mathscr{F})$ asks, in what way might we fail to glue together sections of p-fold intersections satisfying a pth degree agreement condition (that is, a condition on $(p+1)$-fold intersections) into a system of sections on the $(p-1)$-fold intersections.

Theorem 6.3 (Long Exact Cohomology Sequence)**.** *Suppose that*

$$0 \to A \to B \to C \to 0$$

is an exact sequence of chain complexes. Then there is a long exact sequence of cohomology:

$$\begin{tikzcd}
0 \arrow[r] & H^0(A) \arrow[d] & H^1(A) \arrow[d] & H^2(A) \arrow[d] & \cdots \\
& H^0(B) \arrow[d] & H^1(B) \arrow[d] & H^2(B) \arrow[d] & \cdots \\
& H^0(C) \arrow[uur] & H^1(C) \arrow[uur] & H^2(C) \arrow[uur] & \cdots
\end{tikzcd}$$

Proof. We can visualize a short exact sequence of chain complexes in the following way:

$$\begin{tikzcd}
0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
A^0 \arrow[r] \arrow[d] & A^1 \arrow[r] \arrow[d] & A^2 \arrow[r] \arrow[d] & A^3 \arrow[r] \arrow[d] & \cdots \\
B^0 \arrow[r] \arrow[d] & B^1 \arrow[r] \arrow[d] & B^2 \arrow[r] \arrow[d] & B^3 \arrow[r] \arrow[d] & \cdots \\
C^0 \arrow[r] \arrow[d] & C^1 \arrow[r] \arrow[d] & C^2 \arrow[r] \arrow[d] & C^3 \arrow[r] \arrow[d] & \cdots \\
0 & 0 & 0 & 0 &
\end{tikzcd}$$

The diagram is made up of maps of exact sequences of the form

$$\begin{tikzcd}
0 \arrow[r] & A^p \arrow[r] \arrow[d, "d_A^p"] & B_p \arrow[r] \arrow[d, "d_B^p"] & C^p \arrow[r] \arrow[d, "d_C^p"] & 0 \\
0 \arrow[r] & A^{p+1} \arrow[r] & B^{p+1} \arrow[r] & C^{p+1} \arrow[r] & 0
\end{tikzcd}$$

and applying the snake lemma gives the exact sequence:

$$0 \to Z^pA \to Z^pB \to Z^pC \to \frac{A^{p+1}}{dA^p} \to \frac{B^{p+1}}{dB^p} \to \frac{C^{p+1}}{dC^p} \to 0$$

where the Z^p denotes the pth cocycles (the notation of [8, Ch. 1, §1.3]). We can patch together the first half of this sequence for $p-1$ with the second half of the sequence for $p+1$ to obtain:

$$\begin{tikzcd}
 & \frac{A^p}{dA^{p-1}} \arrow[r] \arrow[d, "d_A^p"] & \frac{B^p}{dB^{p-1}} \arrow[r] \arrow[d, "d_B^p"] & \frac{C^p}{dC^{p-1}} \arrow[r] \arrow[d, "d_C^p"] & 0 \\
0 \arrow[r] & Z^{p+1}A \arrow[r] & Z^{p+1}B \arrow[r] & Z^{p+1}C &
\end{tikzcd}$$

Applying the snake lemma again gives

$$H^p(A) \to H^p(B) \to H^p(C) \to H^{p+1}(A) \to H^{p+1}(B) \to H^{p+1}(C)$$

and the proposition follows from patching these sequences together for all p. $\square$

Theorem 6.4. *Let X be a separated scheme and $\mathcal{U} = \{U_i\}$ be an affine open cover. Let*

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \mathscr{F}_3 \to 0$$

be a short exact sequence of quasi-coherent sheaves. Then the induced sequence

$$0 \to C(\mathscr{F}_1) \to C(\mathscr{F}_2) \to C(\mathscr{F}_3) \to 0$$

is a short exact sequence of cochain complexes.

Proof. Because X is separated, every finite intersection $U_{i_0 \cdots i_p}$ is affine. Restricting quasi-coherent sheaves to an affine open keeps them quasi-coherent. On an affine scheme $W = \operatorname{Spec}(A)$, the global sections functor $\Gamma(W, -)$ is exact on quasi-coherent sheaves (equivalently $\mathrm{QCoh}(W) \simeq A$-Mod), so for each $U_{i_0 \cdots i_p}$ we have a short exact sequence:

$$0 \to \mathcal{F}_1(U_{i_0 \cdots i_p}) \to \mathcal{F}_2(U_{i_0 \cdots i_p}) \to \mathcal{F}_3(U_{i_0 \cdots i_p}) \to 0.$$

However by definition the sequence

$$0 \to C(\mathscr{F}_1) \to C(\mathscr{F}_2) \to C(\mathscr{F}_3) \to 0$$

is a direct sum of these sequences, hence exact. □

Now we will relate the Čech cohomology to another more canonical construction.

Definition 6.5 (Sheaf Cohomology)**.** The category of sheaves of modules has *enough injectives*. Therefore the sheaf cohomology can be defined as the right derived functor to the global sections functor $\Gamma(X, -)$. More specifically if $\mathscr{F}$ is a sheaf and

$$0 \to \mathscr{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$$

is an *injective resolution* then the sequence of global sections

$$0 \to \Gamma(X, \mathcal{I}^0) \to \Gamma(X, \mathcal{I}^1) \to \Gamma(X, \mathcal{I}^2) \to \cdots$$

is a cochain complex and cohomology can be defined.

The right derived functor of a left exact functor like $\Gamma(X, -)$ essentially measures how much it fails to be right exact. The reader who is unfamiliar with the notion of a derived functor is encouraged to read that section in [8, Ch. 2, §§2.1–2.5]. Alternatively, one may take results of Theorem 6.6 and Theorem 6.9 as a reasonable black box.

Theorem 6.6. *If X is a separated scheme and $\mathscr{F}$ is a quasi-coherent sheaf on X, then the Čech cohomology of $\mathscr{F}$, computed with respect to any affine open cover, is isomorphic to the sheaf cohomology.*

Proof. The proof is found in [1, Ch. III, §4, Thm. 4.5]. □

Theorem 6.6 is the workhorse result of this section. Essentially, the sheaf cohomology is a very canonical construction and has a lot of nice theoretical properties. On the other hand it is very difficult to actually compute things with. The Čech cohomology is more explicit and allows one to compute by taking a "good" open cover. Therefore a parallel treatment allows one to prove a lot of nontrivial theorems by bouncing these cohomologies off each other.

Corollary 6.7. *If X is an affine scheme and $\mathscr{F}$ a quasi-coherent sheaf then $H^p(X, \mathscr{F}) = \{0\}$ for every $p \geq 1$.*

Proof. We may compute Čech cohomology by Theorem 6.6. Choose the open cover with X as the only set. The theorem follows from the fact that there is no way to choose more than one open set from a single open set. □

Definition 6.8 (Flasque Sheaves)**.** A sheaf $\mathscr{F}$ is *flasque* (or flabby) if the restriction maps $\mathscr{F}(U) \to \mathscr{F}(V)$ are surjective for all open sets $V \subset U$.

Theorem 6.9. *There are the following results on flasque sheaves:*

(a) *All injective objects in the category of sheaves on X are flasque.*
(b) *The global sections functor is exact on flasque sheaves.*
(c) *For any flasque sheaf $\mathscr{F}$ we have $H^p(X, \mathscr{F}) = 0$ for all $p > 0$.*

(d) *The sheaf cohomology may be computed with any flasque resolution. This is a relaxation of the condition requiring an injective resolution.*
(e) *Flasque sheaves are preserved by direct limits.*

Proof. The proof of all these statements may be found in [1, Ch. III, §2]. I will give a proof of (b) since it is equivalent to (c) and (d) by general results on derived functors [8, Ch. 2, §§2.1–2.5]. Furthermore (a) is not necessary for this paper and (e) is a formal verification which can be found in [1, Ch. III, §2] for those who are interested.

Suppose that

$$0 \to \mathscr{F}' \underset{a}{\to} \mathscr{F} \underset{b}{\to} \mathscr{F}'' \to 0$$

with $\mathscr{F}'$ a flasque sheaf. Since the global sections functor is left exact [4, §16, Rem. 16.1(a)] it suffices to show that $\Gamma(U, b)$ is surjective for every open U. Suppose that $s \in \mathscr{F}''(U)$ and consider the set of pairs (V, t) for $V \subseteq U$ open and $t \in \mathscr{F}(V)$ with $b(t)|_V = s|_V$. By exactness of the sequence this set is not empty, and furthermore by Zorn's lemma has an upper bound (V, t). For any $x \in U$ pick a neighborhood $W \subseteq U$ of x and a local lift $t' \in \mathscr{F}(W)$. Note that $t|_{W\cap V} - t'|_{W\cap V}$ lies in $a(\mathscr{F}'(W \cap V))$ after possibly shrinking W. Using flasqueness this difference extends to some $r \in \mathscr{F}'(W)$, therefore replace t' with $t' - a(r)$ so that t and t' agree on $W \cap V$. Now they can be glued into a section on $W \cup V$ and $x \in V$. This shows that $V = U$ as desired. □

The next two theorems are things we would like to be true for a well behaved cohomology theory.

Theorem 6.10 (Grothendieck Vanishing)**.** *Suppose that X is Noetherian of dimension n and $\mathscr{F}$ is a quasi-coherent sheaf. Then $H^p(X, \mathscr{F}) = 0$ for all $p > n$.*

Proof. A full proof may be found in [1, Ch. III, §2, Thm. 2.7]. We only recall the idea. The result is actually topological; on a Noetherian topological space of dimension n, every sheaf has vanishing cohomology above degree n. The proof proceeds by Noetherian induction, repeatedly separating a sheaf into a part supported on an open set and a part supported on a proper closed subset. The closed part has smaller dimension, while the remaining pieces are controlled using flasque resolutions. After at most n such reductions, no higher cohomology can remain. □

Theorem 6.11 (Finiteness of Coherent Cohomology)**.** *If X is proper over k and $\mathscr{F}$ is a coherent sheaf on X, then $H^p(X, \mathscr{F})$ is a finite dimensional k-vector space for every $p \geq 0$ and we denote*

$$h^p(X, \mathscr{F}) := \dim_k H^p(X, \mathscr{F})$$

as its dimension.

Proof. In the projective case, which is enough for the smooth proper curves considered in this paper, this is found in [1, Ch. III, §5, Thm. 5.2]. □

Example 6.12 (Čech Cohomology of $\mathcal{O}(n)$ on $\mathbb{P}^1$)**.** We will compute the cohomology of the twisting sheaves on $\mathbb{P}^1$ using the standard affine cover. Let $U_1 = \operatorname{Spec} k[t]$ and $U_2 = \operatorname{Spec} k[u]$ with $u = t^{-1}$ on the intersection $U_{12} = \operatorname{Spec} k[t, t^{-1}]$. The bundle $\mathcal{O}(n)$ is defined by the transition function t^{-n} on the overlap. This means that a section s_2 on U_2 corresponds to the function $t^n s_2(t^{-1})$ when viewed on U_{12}.

The Čech complex is:

$$C^0(\mathcal{O}(n)) \xrightarrow{d^0} C^1(\mathcal{O}(n)) \xrightarrow{d^1} 0$$

Explicitly, this maps a pair of sections $(P(t), Q(u)) \in k[t] \oplus k[u]$ to:

$$d^0(P, Q) = P(t) - t^n Q(t^{-1}) \in k[t, t^{-1}]$$

Computing H^0: The kernel consists of pairs where $P(t) = t^n Q(t^{-1})$. If $P(t) = \sum a_i t^i$ and $Q(u) = \sum b_j u^j$, equality implies:

$$\sum_{i=0}^{\infty} a_i t^i = \sum_{j=0}^{\infty} b_j t^{n-j}$$

Comparing coefficients, we require $b_j = a_{n-j}$.

- If $n < 0$, the powers of t on the RHS are all negative (since $j \geq 0$), while LHS powers are non-negative. No non-zero solution exists.
- If $n \geq 0$, valid indices are $0 \leq j \leq n$, corresponding to $0 \leq i \leq n$. This gives $n + 1$ independent choices $(1, t, \dots, t^n)$.

Thus $h^0(\mathbb{P}^1, \mathcal{O}(n)) = \max(0, n + 1)$.

Computing H^1: The first cohomology group is the cokernel of d^0. Recall that $\{t^k \mid k \in \mathbb{Z}\}$ is a basis for $k[t, t^{-1}]$. The image of d^0 contains all t^k for $k \geq 0$ (from $P(t)$) and all t^k for $k \leq n$ (from $t^n Q(t^{-1})$).

- If $n \geq -1$, the union of these sets is all integers k, so the map is surjective and $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$.
- If $n < -1$, the integers strictly between n and 0 are not in the image. The missing monomials are $t^{n+1}, t^{n+2}, \dots, t^{-1}$. The number of such monomials is $(-1) - (n + 1) + 1 = -n - 1$.

Thus $h^1(\mathbb{P}^1, \mathcal{O}(n)) = \max(0, -n - 1)$.

Theorem 6.13 (Serre Duality on $\mathbb{P}^1$)**.** *For every $n \in \mathbb{Z}$, the residue pairing*

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \times H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1} \otimes \mathcal{O}(n)^\vee) \to k$$

is perfect.

Proof. By Theorem 5.15 and Construction 4.9,

$$\Omega_{\mathbb{P}^1} \otimes \mathcal{O}(n)^\vee \simeq \mathcal{O}(-2) \otimes \mathcal{O}(-n) \simeq \mathcal{O}(-n-2).$$

Thus the desired pairing is:

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \times H^1(\mathbb{P}^1, \mathcal{O}(-n-2)) \to k$$

which is given by the residue pairing: By the computation above:

$$h^0(\mathbb{P}^1, \mathcal{O}(n)) = \max(0, n+1) \qquad h^1(\mathbb{P}^1, \mathcal{O}(-n-2)) = \max(0, n+1)$$

If $n < 0$, both vector spaces are zero, so there is nothing to prove.

Now suppose that $n \geq 0$. Let e be the standard frame for $\mathcal{O}(n)$ on $U_0 = \operatorname{Spec} k[x]$, and let $e^\vee$ be the dual frame for $\mathcal{O}(n)^\vee$. Then $H^0(\mathbb{P}^1, \mathcal{O}(n))$ has basis

$$e, xe, \dots, x^n e,$$

while $H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1} \otimes \mathcal{O}(n)^\vee)$ is represented by the principal parts

$$x^{-1} dx \otimes e^\vee, \ x^{-2} dx \otimes e^\vee, \ \dots, \ x^{-n-1} dx \otimes e^\vee.$$

We define the residue pairing to take the -1 coefficient of the product. Therefore

$$\operatorname{res}_0\left((x^i e)(x^{-j-1}dx \otimes e^\vee)\right) = \operatorname{res}_0(x^{i-j-1}dx) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

so the two bases are dual to each other and the residue pairing is perfect. $\square$

This is the simplest example of the general Serre duality theorem that is one of the overarching goals of this paper.

Definition 6.14 (Euler Characteristic)**.** Suppose that X satisfies the hypotheses of Theorem 6.10 and Theorem 6.11. The *Euler characteristic* is the alternating sum of the dimensions of the cohomology groups

$$\chi(X, \mathscr{F}) = \sum_{p=0}^{\infty} (-1)^p h^p(X, \mathscr{F})$$

which is well defined by Theorem 6.10 and Theorem 6.11

This definition should be thought of as somewhat analogous to the famous equation for a three dimensional polyhedron

$$\chi = V - E + F$$

which states that the Euler characteristic equals the number of vertices minus edges plus faces. This reduces a very complicated topological idea to a matter of counting.

Theorem 6.15 (Additivity of Euler Characteristic)**.** *If*

$$0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \mathscr{F}_3 \to 0$$

is an exact sequence of sheaves then:

$$\chi(\mathscr{F}_2) = \chi(\mathscr{F}_1) + \chi(F_3)$$

Proof. Take the long exact sequence of cohomology

$$\begin{tikzcd}
0 \arrow[r] & H^0(\mathscr{F}_1) \arrow[d] & H^1(\mathscr{F}_1) \arrow[d] & H^2(\mathscr{F}_1) \arrow[d] & \cdots \\
& H^0(\mathscr{F}_2) \arrow[d] & H^1(\mathscr{F}_2) \arrow[d] & H^2(\mathscr{F}_2) \arrow[d] & \cdots \\
& H^0(\mathscr{F}_3) \arrow[uur] & H^1(\mathscr{F}_3) \arrow[uur] & H^2(\mathscr{F}_3) \arrow[uur] & \cdots
\end{tikzcd}$$

and note since $\dim_k$ is additive the alternating sum of the dimensions of the homologies along this sequence is 0. This sum is exactly the alternating sum of the Euler characteristics as desired. $\square$

This additivity is the reason our cheap Riemann-Roch theorem is true.

Definition 6.16 (Genus)**.** The genus is defined as the dimension of the first cohomology group of the structure sheaf:

$$g = h^1(X, \mathcal{O}_X)$$

In general, the genus says something about the number of holes a topological space has. For instance a sphere has genus 0 but a torus has genus 1.

Remark 6.17. In the case of a smooth proper curve

$$\chi(X, \mathscr{F}) = h^0(X, \mathscr{F}) - h^1(X, \mathscr{F})$$

by Theorem 6.10.

Finally, we can prove the Riemann-Roch theorem. As before let C be a smooth proper curve.

Theorem 6.18 (Cheap Riemann-Roch)**.** *If L is a line bundle over C then*

$$h^0(L) - h^1(L) = \deg L + 1 - g$$

where $h^p(L) := h^p(C, L)$.

Proof. Consider the line bundle $L_1 \otimes L_2$ where $L_2 = \mathcal{L}^D$ for some effective divisor D. Now consider the exact sequence in Theorem 4.10

$$0 \to L_1 \to L_1 \otimes L_2 \to Q \to 0$$

Using Theorem 6.15 we obtain:

$$\chi(L_1 \otimes L_2) = \chi(L_1) + \chi(Q)$$

However note that $h^0(Q) = \deg L_2$ by construction and $h^p(Q) = 0$ for $p > 0$ since Q is flasque. Therefore $\chi(Q) = \deg L_2$ and we obtain

$$\chi(L_1 \otimes L_2) = \chi(L_1) + \deg L_2 \tag{1}$$

Now take a "common ancestor" L' of L and $\mathcal{O}_C$ with

$$L = L' \otimes \mathcal{L}^{D_1} \qquad \mathcal{O}_C = L' \otimes \mathcal{L}^{D_2}$$

for effective divisors D_1 and D_2. By applying Eq. (1) to L and $\mathcal{O}_C$ and comparing it follows that

$$\chi(L) = \chi(\mathcal{O}_C) + \deg L$$

which is the statement that the Euler characteristic is additive in degree. Now use Remark 6.17 to rewrite the LHS and RHS as

$$h^0(L) - h^1(L) \qquad h^0(\mathcal{O}_C) - h^1(\mathcal{O}_C) + \deg L = \deg L + 1 - g$$

respectively, using definitions and $h^0(\mathcal{O}_C) = 1$. This is the desired equation. □

7. Serre Duality and Integration

You may notice that the canonical divisor does not appear in the above formulation of Riemann-Roch. This is because we have not yet proven the full theorem. In order to achieve the desired form and draw connections to the canonical divisor, we need to invoke a deep result known as *Serre duality*. Serre duality can be thought of as a statement about integration in a sense that will become more clear as we proceed. A full algebraic proof with all the diagram chases may be found in [1, Ch. III, §7] or [10].

For the remainder L will denote a line bundle on C. The first step is to reinterpret cohomology in terms of principal parts.

Construction 7.1. Using the language of rational sections from Example 4.4, define a sheaf L_{rat} by:

$$L_{\text{rat}}(U) = \begin{cases} L \otimes_{\mathcal{O}_C} K(C), & U \neq \emptyset \\ 0, & U = \emptyset \end{cases}$$

Since the restriction maps on nonempty opens are the identity map, L_{rat} is flasque. Now define the sheaf of principal parts of L by the exact sequence

$$0 \to L \to L_{\mathrm{rat}} \to \mathrm{Prin}(L) \to 0$$

Since L_{rat} is flasque, applying cohomology gives an exact sequence:

$$0 \to H^0(C, L) \to L \otimes_{\mathcal{O}_C} K(C) \to \Gamma(C, \mathrm{Prin}(L)) \to H^1(C, L) \to 0$$

Therefore we may write:

$$H^1(C, L) \cong \frac{\Gamma(C, \mathrm{Prin}(L))}{\mathrm{pp}(L \otimes_{\mathcal{O}_C} K(C))}$$

Remark 7.2. The name principal parts comes from the following local picture. If $P \in C$, choose a local trivialization e of L near P and a uniformizer $t \in \mathcal{O}_{C,P}$. Then a rational section of L can be written locally as a Laurent expression

$$s = \left(\sum_{i \gg -\infty} a_i t^i \right) e$$

Passing to $\mathrm{Prin}(L)_P$ means quotienting by the regular local sections of L. Thus all nonnegative powers of t are forgotten, and only the polar part

$$\left(a_{-m} t^{-m} + \cdots + a_{-1} t^{-1} \right) e$$

remains. This finite Laurent tail is the principal part of the rational section at P. Although this description uses choices, the quotient sheaf

$$\mathrm{Prin}(L) = L_{\mathrm{rat}} / L$$

is intrinsic.

In this way, a global section of $\mathrm{Prin}(L)$ is a finite collection of prescribed local principal parts. The *Mittag-Leffler problem* asks whether these local principal parts come from a global rational section of L. In the exact sequence

$$0 \to H^0(C, L) \to L \otimes_{\mathcal{O}_C} K(C) \to \Gamma(C, \mathrm{Prin}(L)) \to H^1(C, L) \to 0$$

the image of a collection of principal parts in $H^1(C, L)$ is exactly the obstruction to solving this problem. Thus the prescribed principal parts can be realized by a global rational section if and only if their class in $H^1(C, L)$ is zero.

The analytic analogue to this construction is the Laurent tail divisor in [12, Ch. VI, §2].

Construction 7.3. Suppose we plug in Ω_C for L, choose a rational differential ω, and a point $P \in C$. Since C is smooth, we may choose a uniformizing parameter t at P and write:

$$\omega = \left(\sum_{n \gg -\infty} a_n t^n \right) dt$$

We define the residue of ω at P to be:

$$\mathrm{res}_P(\omega) = a_{-1}$$

A local calculation shows that this definition is independent of the choice of uniformizer.

Lemma 7.4. *Let $f : C \to C'$ be a finite morphism of smooth curves over an algebraically closed field k, and let $Q \in C'$. Choose a uniformizer t at Q and, for each $P \in f^{-1}(Q)$, a uniformizer u_P at P. Then*

$$\widehat{\mathcal{O}}_{C',Q} \simeq k[[t]], \qquad \widehat{\mathcal{O}}_{C,P} \simeq k[[u_P]],$$

and

$$(\widehat{f_*\mathcal{O}_C})_Q \simeq \prod_{P \in f^{-1}(Q)} \widehat{\mathcal{O}}_{C,P}.$$

Consequently,

$$K(C) \otimes_{K(C')} k((t)) \simeq \prod_{P \in f^{-1}(Q)} k((u_P)).$$

is a product of finite separable extensions of local fields.

Proof. Since C and C' are smooth curves, the local rings $\mathcal{O}_{C,P}$ and $\mathcal{O}_{C',Q}$ are discrete valuation rings. Their residue fields are k, so the chosen uniformizers induce isomorphisms

$$\widehat{\mathcal{O}}_{C',Q} \simeq k[[t]], \qquad \widehat{\mathcal{O}}_{C,P} \simeq k[[u_P]].$$

Set $A = \mathcal{O}_{C',Q}$ and $B = (f_*\mathcal{O}_C)_Q$. Since B is finite over A, completion gives

$$B \otimes_A \widehat{A} \simeq \prod_{P \in f^{-1}(Q)} \widehat{B_P} = \prod_{P \in f^{-1}(Q)} \widehat{\mathcal{O}}_{C,P}.$$

Since completion commutes with finite modules, the left-hand side is $\widehat{B} = (\widehat{f_*\mathcal{O}_C})_Q$. Localizing t now gives

$$K(C) \otimes_{K(C')} k((t)) \simeq \prod_{P \in f^{-1}(Q)} k((u_P)),$$

as desired. $\square$

Lemma 7.5 (Trace and Residues)**.** *Let $f : C \to C'$ be a finite morphism of smooth proper curves and let ω be a rational differential on C. Then for every point $Q \in C'$:*

$$\operatorname{res}_Q(\operatorname{Tr}_{C/C'} \omega) = \sum_{P \in f^{-1}(Q)} \operatorname{res}_P(\omega)$$

where $\operatorname{Tr}_{C/C'}$ denotes the trace map on rational differentials.

Proof. We prove this after completing the local rings at the points in question. By Lemma 7.4, the extension of local fields decomposes as:

$$K(C) \otimes_{K(C')} k((t)) \cong \prod_{P \in f^{-1}(Q)} k((u_P))$$

The field trace is compatible with this product decomposition, so it suffices to prove the statement for a finite extension:

$$k((t)) \subseteq k((u))$$

In this case both sides are k-linear functionals on rational differentials over $k((u))$. They vanish on exact differentials, since the residue of a derivative is 0 and trace commutes with differentiation. Therefore it suffices to check the claim on one differential with nonzero residue.

Let e be the ramification index, so that $\mathrm{ord}_u(t) = e$. Then:

$$\mathrm{res}_u\left(\frac{dt}{t}\right) = e$$

On the other hand

$$\mathrm{Tr}_{k((u))/k((t))}\left(\frac{dt}{t}\right) = \mathrm{Tr}_{k((u))/k((t))}(1)\frac{dt}{t} = e\frac{dt}{t}$$

because the residue fields are all equal to k. Hence

$$\mathrm{res}_t\left(\mathrm{Tr}_{k((u))/k((t))}\left(\frac{dt}{t}\right)\right) = \mathrm{res}_t\left(e\frac{dt}{t}\right) = e$$

and the two residue functionals agree locally. Summing over the points above Q gives the desired formula. □

The above theorem tells us that this trace operation corresponds geometrically to taking the pushforward of a differential form. Using this, we can prove the residue theorem by pushing a form down to $\mathbb{P}^1$, where we can do explicit computations.

Theorem 7.6 (Algebraic Residue Theorem)**.** *Let C be a smooth proper curve and let ω be a rational differential on C. Then:*

$$\sum_{P\in C} \mathrm{res}_P(\omega) = 0$$

Proof. We first prove the theorem in the case $C = \mathbb{P}^1$. Let x be the affine coordinate on $\mathbb{P}^1$. Then every rational differential on $\mathbb{P}^1$ can be written as

$$\eta = h(x)dx$$

for some rational function $h(x) \in k(x)$. By partial fractions we may write

$$h(x) = p(x) + \sum_{a\in k}\sum_{m\geq 1} \frac{c_{a,m}}{(x-a)^m}$$

and compute the contribution from $\frac{c_{a,m}}{(x-a)^m}$ to the residues:

- If $m \neq 1$ then there is no residue.
- If $m = 1$ then there is residue $c_{a,m}$ and a and $-c_{a,m}$ at ∞.

Therefore the theorem is true for $\mathbb{P}^1$.

Now let C be a general smooth proper curve. By Noether normalization choose a nonconstant rational function $x \in K(C)$ such that $K(C)$ is a finite separable extension of $k(x)$. This gives a finite morphism

$$f : C \to \mathbb{P}^1.$$

Since dx is a basis for the rational differentials after passing to $K(C)$, we may write

$$\omega = a\,dx$$

for some $a \in K(C)$. The trace of ω is defined by

$$\mathrm{Tr}_{C/\mathbb{P}^1}(\omega) = \mathrm{Tr}_{K(C)/k(x)}(a)dx$$

which is a rational differential on $\mathbb{P}^1$. Using Lemma 7.5, we get:

$$\sum_{P\in C}\mathrm{res}_P(\omega) = \sum_{Q\in\mathbb{P}^1}\sum_{P\in f^{-1}(Q)}\mathrm{res}_P(\omega) = \sum_{Q\in\mathbb{P}^1}\mathrm{res}_Q(\mathrm{Tr}_{C/\mathbb{P}^1}\,\omega)$$

But $\mathrm{Tr}_{C/\mathbb{P}^1}\,\omega$ is a rational differential on $\mathbb{P}^1$, so by the computation above,

$$\sum_{Q\in\mathbb{P}^1} \mathrm{res}_Q(\mathrm{Tr}_{C/\mathbb{P}^1}\,\omega) = 0 \implies \sum_{P\in C} \mathrm{res}_P(\omega) = 0$$

as desired. □

The above theorem is an exact algebraic analogue to the statement that the sum of residues of a meromorphic differential on a compact Riemann surface is 0. This justifies why we can descend the residue pairing to cohomology.

Definition 7.7. Define an algebraic integration map

$$\int_C : H^1(C, \Omega_C) \to k$$

by representing a class in $H^1(C, \Omega_C)$ some $\omega_P \in \mathrm{Prin}(\Omega_C)$ Then set

$$\int_C [(\omega_P)_P] := \sum_{P\in C} \mathrm{res}_P(\omega_P).$$

This is well defined for two reasons. Firstly, adding a regular differential to ω_P does not change the residue at P. Secondly, changing the representative by the principal parts of a global rational differential changes the sum by the total residue of that differential, which is 0 by the residue theorem. Thus the map $\int_C$ is canonical. This is the algebraic replacement for integration over the curve.

Definition 7.8 (Residue Pairing)**.** By Construction 4.9 there is a canonical evaluation map

$$\mathrm{ev} : L \otimes_{\mathcal{O}_C} L^\vee \longrightarrow \mathcal{O}_C.$$

Tensoring with Ω_C, we obtain a canonical multiplication map

$$L \otimes_{\mathcal{O}_C} (\Omega_C \otimes_{\mathcal{O}_C} L^\vee) \longrightarrow \Omega_C \qquad \ell \otimes (\omega \otimes \lambda) \longmapsto \lambda(\ell)\omega.$$

Thus a section of L can multiply an $L^\vee$-valued differential to produce an ordinary differential. Let

$$s \in H^0(C, L), \qquad \xi \in H^1(C, \Omega_C \otimes L^\vee)$$

and consider the diagram

$$\begin{tikzcd}
0 \arrow[r] & \Omega_C \otimes L^\vee \arrow[r] \arrow[d, "\cdot s"] & (\Omega_C \otimes L^\vee)_{\mathrm{rat}} \arrow[r] \arrow[d, "\cdot s"] & \mathrm{Prin}(\Omega_C \otimes L^\vee) \arrow[r] \arrow[d, dashed, "\exists!"] & 0 \\
0 \arrow[r] & \Omega_C \arrow[r] & (\Omega_C)_{\mathrm{rat}} \arrow[r] & \mathrm{Prin}(\Omega_C) \arrow[r] & 0
\end{tikzcd}$$

which extends the multiplication map to:

$$H^0(C, L) \times \mathrm{Prin}(\Omega_C \otimes L^\vee) \longrightarrow \mathrm{Prin}(\Omega_C)$$

Using Construction 7.1, represent ξ by a global principal part $\alpha \in \Gamma(C, \mathrm{Prin}(\Omega_C \otimes L^\vee))$ and define

$$\langle s, \xi\rangle = \int_C s\alpha$$

to be the residue pairing. Locally, if near P we choose a local frame e for L, with dual frame $e^\vee$, and write

$$s = fe, \qquad \alpha_P = \left[\left(\sum_{n\gg-\infty} a_n t^n\right) dt \otimes e^\vee\right],$$

then

$$s\alpha_P = \left[f \left(\sum_{n \gg -\infty} a_n t^n \right) dt \right] \in \mathrm{Prin}(\Omega_C)_P.$$

and the residue pairing is:

$$\langle s, \xi \rangle := \int_C s\xi := \sum_{P \in C} \mathrm{res}_P(s\alpha_P)$$

This is independent of the chosen representative of ξ. Indeed, changing α by a global rational section of $\Omega_C \otimes L^\vee$ changes $s\alpha$ by a global rational differential, whose total residue is zero by the residue theorem.

This is the residue formulation of Serre duality. It says that we pair a section with a dual-valued differential, obtain an ordinary differential, and then integrate by summing residues.

Remark 7.9. Note that if $\xi \in H^1(C, \Omega_C \otimes L^\vee)$ is the 0 class, then it is clear that

$$\langle s, \xi \rangle = 0 \qquad \text{for all } s \in H^0(L).$$

Serre duality will tell us that the map

$$H^1(\Omega_C \otimes L^\vee) \to H^0(L)^\vee \qquad \xi \longmapsto \langle -, \xi \rangle$$

is an isomorphism. This gives us the converse statement, that is, if ξ pairs to 0 with every element of $H^0(L)$, then it is 0 in H^1, meaning we can solve the Mittag-Leffler problem for this class of principal parts.

Lemma 7.5 with $C' = \mathbb{P}^1$ says that, after applying trace, the total residue of a rational differential on C can be computed downstairs on $\mathbb{P}^1$. Thus the residue pairing on C should become the ordinary residue pairing on $\mathbb{P}^1$ after pushing forward along f. Since the Serre-dual partner of f_*L on $\mathbb{P}^1$ is $\Omega_{\mathbb{P}^1} \otimes (f_*L)^\vee$, we are led to expect that the pushed-forward sheaf $f_*(\Omega_C \otimes L^\vee)$ is identified with this dual bundle via the trace pairing. This is the following theorem.

Lemma 7.10. *Let $f : C \to \mathbb{P}^1$ be a finite separable morphism of smooth proper curves, and let L be a line bundle on C. Then the trace pairing*

$$f_*(\Omega_C \otimes L^\vee) \otimes f_*(L) \to \Omega_{\mathbb{P}^1} \qquad \eta \otimes s \longmapsto \mathrm{Tr}_{C/\mathbb{P}^1}(\eta s)$$

is perfect.

Proof. The pairing is defined by multiplying local sections on C and then applying the trace on differentials:

$$f_*(\Omega_C \otimes L^\vee) \otimes f_*L \longrightarrow \Omega_{\mathbb{P}^1} \qquad \eta \otimes s \longmapsto \mathrm{Tr}_{C/\mathbb{P}^1}(\eta s)$$

We show that this pairing is perfect.

Perfectness is local on $\mathbb{P}^1$, so fix a point $Q \in \mathbb{P}^1$. By Lemma 7.4 after completing at Q, the finite algebra decomposes as a product over the points $P \in f^{-1}(Q)$. Since completion is faithfully flat, it is enough to consider one such point P. Set

$$R = \widehat{\mathcal{O}}_{\mathbb{P}^1, Q} \qquad S = \widehat{\mathcal{O}}_{C,P}$$

where $R \subseteq S$ is a finite separable extension of complete discrete valuation rings. Choose uniformizers $t \in R$ and $u \in S$. By Theorem 5.8 and Example 5.7, the completed stalks of the modules of differentials are:

$$\widehat{\Omega}_{\mathbb{P}^1, Q} = R\, dt \qquad \widehat{\Omega}_{C,P} = S\, du$$

Since L is trivial over the local ring S, choosing a local frame of L and its dual reduces the pairing to:

$$S \times S\,du \longrightarrow R\,dt \qquad (a, b\,du) \longmapsto \mathrm{Tr}_{S/R}(ab\,du)$$

There is a unique $\delta \in S$ such that:

$$dt = \delta\,du$$

Completing the first exact sequence for Kähler differentials Lemma 5.5 gives

$$S\,dt \longrightarrow S\,du \longrightarrow \widehat{\Omega}_{S/R} \longrightarrow 0$$

The first map is multiplication by δ, so

$$\widehat{\Omega}_{S/R} \simeq S/(\delta)$$

Thus the different of S/R is $\mathrm{Ann}_S(\Omega_{S/R}) = (\delta)$ by [13, Ch. III, §2, Def. 2.1 and Prop. 2.7]. Equivalently, its inverse different is:

$$\mathfrak{D}_{S/R}^{-1} = \{x \in \mathrm{Frac}(S) : \mathrm{Tr}_{S/R}(xS) \subseteq R\} = \delta^{-1}S$$

Now let $b\,du$ be a rational differential. Since

$$b\,du = b\delta^{-1}\,dt,$$

the definition of the trace on differentials gives

$$\mathrm{Tr}_{S/R}(ab\,du) = \mathrm{Tr}_{S/R}(ab\delta^{-1})\,dt.$$

Therefore $b\,du$ pairs into $R\,dt$ with every $a \in S$ if and only if

$$\mathrm{Tr}_{S/R}(ab\delta^{-1}) \in R \qquad \text{for every } a \in S.$$

By the definition of the inverse different, this is equivalent to

$$b\delta^{-1} \in \mathfrak{D}_{S/R}^{-1} = \delta^{-1}S,$$

and hence to $b \in S$. Thus the differentials which pair integrally with every element of S are exactly the regular differentials $S\,du$.

The trace pairing is nondegenerate over the fraction fields because the extension is separable. The calculation above therefore shows that $S\,du$ is precisely the dual lattice of S with values in $R\,dt$. Therefore the pairing is perfect as desired. □

Geometrically, the finite map f lets us view a section of L as a collection of values on the points lying over each point of $\mathbb{P}^1$. A section of $\Omega_C \otimes L^\vee$ then tests these values pointwise, and the trace sums the resulting differential contributions into a differential downstairs on $\mathbb{P}^1$. The perfectness says that this fibrewise integration loses no information, even at ramified points, because Ω_C contains exactly the correction needed to make the trace-dual of f_*L equal to $f_*(\Omega_C \otimes L^\vee)$.

Lemma 7.11. *Let E be a vector bundle on $\mathbb{P}^1$. Then the residue pairing gives a perfect pairing:*

$$H^0(\mathbb{P}^1, E) \times H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1} \otimes E^\vee) \to k$$

Proof. By the Birkhoff-Grothendieck theorem [1, Ch. V, Ex. 2.6], every vector bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles:

$$E \simeq \bigoplus_i \mathcal{O}_{\mathbb{P}^1}(n_i)$$

The residue pairing is compatible with direct sums, so it suffices to prove the result for $E = \mathcal{O}_{\mathbb{P}^1}(n)$. This is Theorem 6.13. □

Theorem 7.12 (Serre Duality)**.** *For every line bundle L over C there is a natural perfect pairing:*

$$H^0(C, L) \times H^1(C, \Omega_C \otimes L^\vee) \to k$$

Proof. Choose a nonconstant separable rational function $x \in K(C)$. This gives a finite separable morphism:

$$f : C \to \mathbb{P}^1$$

Since f is finite, the pushforward functor f_* is exact. Therefore

$$H^i(C, L) \simeq H^i(\mathbb{P}^1, f_*L)$$

for every i. Moreover, since C and $\mathbb{P}^1$ are smooth curves, f_*L is a vector bundle on $\mathbb{P}^1$. The trace pairing of Lemma 7.10 gives the identification:

$$f_*(\Omega_C \otimes L^\vee) \simeq (f_*L)^\vee \otimes \Omega_{\mathbb{P}^1}$$

therefore

$$H^1(C, \Omega_C \otimes L^\vee) \simeq H^1(\mathbb{P}^1, f_*(\Omega_C \otimes L^\vee)) \simeq H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1} \otimes (f_*L)^\vee)$$

and under these identifications, the pairing on C becomes the residue pairing

$$H^0(\mathbb{P}^1, f_*L) \times H^1(\mathbb{P}^1, \Omega_{\mathbb{P}^1} \otimes (f_*L)^\vee) \to k$$

which is perfect by Lemma 7.11. Hence the original pairing on C is perfect.

It remains only to check that this is the same pairing as the one defined by residues on C. Let $s \in H^0(C, L)$, and let α be a principal-part representative of a class in

$$H^1(C, \Omega_C \otimes L^\vee)$$

so the pairing on C is

$$\sum_{P \in C} \operatorname{res}_P(s\alpha_P)$$

and after pushing forward to $\mathbb{P}^1$, the corresponding expression is:

$$\sum_{Q \in \mathbb{P}^1} \operatorname{res}_Q \left(\operatorname{Tr}_{C/\mathbb{P}^1}(s\alpha)\right)$$

These two expressions are equal by the compatibility of trace and residues from Lemma 7.5 as desired □

Remark 7.13 (Integration)**.** When $k = \mathbb{C}$, the smooth complex points of C form a compact Riemann surface. In this setting a class in $H^1(C, L)$ can be represented analytically by a smooth L-valued $(0,1)$-form η, while a section of $H^0(C, \Omega_C \otimes L^\vee)$ is a holomorphic $L^\vee$-valued $(1,0)$-form ω. Multiplying them gives an ordinary $(1,1)$-form, and the Serre pairing is:

$$\langle \omega, \eta \rangle = \int_C \omega \wedge \eta$$

If η is changed by a $\bar{\partial}$-exact form, the integral changes by the integral of an exact form, which is 0 by Stokes' theorem. This is the analytic reason that the pairing descends to cohomology.

The residue description above is the algebraic manifestation of the same operation. A meromorphic Čech representative can be converted into a smooth Dolbeault representative by a partition of unity. Then Stokes' theorem reduces the integral of the resulting $(1, 1)$-form to small contour integrals around the poles. Those contour integrals are exactly residues:

$$\frac{1}{2\pi i} \int_{\gamma_P} \omega = \operatorname{res}_P(\omega)$$

Thus the formula

$$\langle s, \xi \rangle = \sum_{P \in C} \operatorname{res}_P(s\alpha_P)$$

is not just an analogy with integration. For complex curves, it is what integration becomes after translating the problem into algebraic principal parts.

8. The Riemann-Roch Theorem

We now have all the abstract machinery required to prove the Riemann-Roch theorem. We will also see that this theorem is powerful enough to quickly prove a lot of other strong theorems, illustrating its importance. In particular, we will finally see that many of the algebraic notions we have defined behave like the topological or geometrical notions they are named after.

Theorem 8.1 (Riemann-Roch)**.** *For every line bundle L on C:*

$$h^0(C, L) - h^0(C, \Omega_C \otimes L^\vee) = \deg L + 1 - g$$

Proof. Cheap Riemann-Roch (Theorem 6.18) gives

$$h^0(C, L) - h^1(C, L) = \deg L + 1 - g$$

By Serre duality applied to $\Omega_C \otimes L^\vee$,

$$H^1(C, L)^\vee \cong H^0(C, \Omega_C \otimes L^\vee)$$

so replacing $h^1(C, L)$ by $h^0(C, \Omega_C \otimes L^\vee)$ proves the theorem. □

Corollary 8.2 (The canonical degree and the genus)**.** *For every canonical divisor K_C*

$$\deg K_C = 2g - 2.$$

and the following algebraic definitions of the genus agree:

$$g = h^1(C, \mathcal{O}_C) = 1 - \chi(C, \mathcal{O}_C) = h^0(C, \Omega_C) = 1 + \frac{1}{2} \deg K_C$$

Proof. Serre duality applied to Ω_C gives $h^0(C, \Omega_C) = h^1(C, \mathcal{O}_C) = g$, while Serre duality applied to $\mathcal{O}_C$ gives $h^1(C, \Omega_C) = h^0(C, \mathcal{O}_C) = 1$. Applying cheap Riemann-Roch to $L = \Omega_C$ gives

$$g - 1 = h^0(C, \Omega_C) - h^1(C, \Omega_C) = \deg \Omega_C + 1 - g$$

so $\deg K_C = \deg \Omega_C = 2g - 2$. Since $h^0(C, \mathcal{O}_C) = 1$, we also have $1 - \chi(C, \mathcal{O}_C) = h^1(C, \mathcal{O}_C)$, and all the displayed definitions agree. □

Theorem 8.3 (Riemann-Hurwitz)**.** *Let $f : C \to C'$ be a finite separable morphism of smooth proper curves, and set $d = \deg f$. Define:*

$$R_f := \sum_{P \in C} \ell_{\mathcal{O}_{C,P}}\big((\Omega_{C/C'})_P\big) P$$

Then:

$$2\chi(C, \mathcal{O}_C) = 2d\chi(C', \mathcal{O}_{C'}) - \deg R_f$$

If f is tamely ramified, then $R_f = \sum_{P \in C}(e_P - 1)P$, so:

$$2\chi(C, \mathcal{O}_C) = 2d\chi(C', \mathcal{O}_{C'}) - \sum_{P \in C}(e_P - 1)$$

Proof. The exact sequence for Kähler differentials gives

$$0 \to f^*\Omega_{C'} \to \Omega_C \to \Omega_{C/C'} \to 0$$

where the first map is injective because f is separable. By additivity of Euler characteristic:

$$\chi(C, \Omega_C) = \chi(C, f^*\Omega_{C'}) + \deg R_f$$

By Corollary 8.2, $\chi(C, \Omega_C) = -\chi(C, \mathcal{O}_C)$ and $\deg \Omega_{C'} = -2\chi(C', \mathcal{O}_{C'})$. Applying cheap Riemann-Roch to the line bundle $f^*\Omega_{C'}$ on C gives

$$\chi(C, f^*\Omega_{C'}) = \deg(f^*\Omega_{C'}) + \chi(C, \mathcal{O}_C) = d \deg \Omega_{C'} + \chi(C, \mathcal{O}_C)$$

therefore

$$-\chi(C, \mathcal{O}_C) = -2d\chi(C', \mathcal{O}_{C'}) + \chi(C, \mathcal{O}_C) + \deg R_f$$

which rearranges to the desired formula. Finally, in the tame case, if $Q = f(P)$ and t, u are uniformizers at Q, P, then locally $f^*t = au^{e_P}$ for a unit a. Since e_P is nonzero in k, the differential $d(f^*t)$ vanishes to order $e_P - 1$, so $\ell_{\mathcal{O}_{C,P}}((\Omega_{C/C'})_P) = e_P - 1$. □

Theorem 8.4 (Topological and algebraic Euler characteristic)**.** *Suppose $k = \mathbb{C}$, and let $C(\mathbb{C})$ be the compact Riemann surface associated to C. Then*

$$\chi_{\text{top}}(C(\mathbb{C})) = 2\chi(C, \mathcal{O}_C)$$

in particular the topological genus of $C(\mathbb{C})$ agrees with the algebraic genus $h^1(C, \mathcal{O}_C)$.

Proof. We may assume the statement for $\mathbb{P}^1$. Choose a nonconstant rational function on C, giving a finite morphism:

$$f : C \to \mathbb{P}^1$$

Since we are over $\mathbb{C}$, this morphism is separable and tamely ramified. The topological Riemann-Hurwitz formula gives:

$$\chi_{\text{top}}(C(\mathbb{C})) = d\chi_{\text{top}}(\mathbb{P}^1(\mathbb{C})) - \sum_{P \in C}(e_P - 1)$$

By assumption, $\chi_{\text{top}}(\mathbb{P}^1(\mathbb{C})) = 2\chi(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1})$. Therefore

$$\chi_{\text{top}}(C(\mathbb{C})) = 2d\chi(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) - \sum_{P \in C}(e_P - 1)$$

which is exactly the algebraic Riemann-Hurwitz formula from Theorem 8.3 applied to $f : C \to \mathbb{P}^1$. Hence:

$$\chi_{\text{top}}(C(\mathbb{C})) = 2\chi(C, \mathcal{O}_C).$$

Since $\chi(C, \mathcal{O}_C) = 1 - h^1(C, \mathcal{O}_C)$, this is equivalent to saying that the topological and algebraic genera agree. □

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